



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 25- 401 - 1

Date: 29-09-2020

EDMS 25-401

SPECIFICATION

FOR

ELECTRICAL INSULATING OIL

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1- SCOPE

This specification covers the minimum technical requirements for materials, testing and packing of the electrical insulating oil to be used in transformers, switchgears, voltage regulators and other electrical equipments. The oil shall be suitable for use in both 12 kV and 24 kV systems of the distribution network of Electricity Distribution Company (___ EDC).

2- APPLICABLE CODES and STANDARDS

The oil covered in this specification shall comply with the latest edition/amendment of the following codes and standards. Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply.

- IEC 60296: Fluids for electrotechnical applications - Specification for Unused Mineral Insulating Oils for Transformers and Switchgear.
- IEEE C57-106: Guide for Acceptance and Maintenance of Insulating Mineral Oil in Electrical Equipment
- ASTM D3487: Specification for Mineral Insulating Oils Used in Electrical Apparatus.
- ASTM 117: Testing and Specifications for Electrical Insulating Oils of Petroleum Origin.
- IEC 60475: Method of sampling insulating liquids
- ASTM D 923: Sampling Electrical Insulating Liquids.



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- IEC 60422: Mineral insulating oils in electrical equipment - Supervision and maintenance guidance
- IEC 60156: Insulating liquids - Determination of the breakdown voltage at power frequency - Test method
- IEC 60247: Insulating liquids - Measurement of relative permittivity, dielectric dissipation factor (tan d) and d.c. resistivity
- IEC 61125: Insulating liquids - Test methods for oxidation stability - Test method for evaluating the oxidation stability of insulating liquids in the delivered state.

3- FEATURES

- The oil shall be hydrocarbon mineral oil specifically manufactured for use as electrical insulating oil.
- The oil can be mixed with other equivalent transformer oils by any quantity without any changes affecting their efficiency or validity for insulation inside the electrical equipments.

4- TECHNICAL CHARACTERISTICS

- The oil shall meet the requirements and the technical characteristics indicated in Table (1).



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Table (1)

Item	Specification	Unit	Limit	Test Method
1	Viscosity at 40 °C	mm ² / s	≤ 12	ISO 3104
2	Flash point	°C	> 140	ISO 2719
3	Pour point	°C	≤ -30	ISO 3016
4	Density at 20 °C	kg / m ³	≤ 895	ISO 3675, ISO 12185
5	Dissipation factor at 90 °C	---	< 0.005	IEC 60247, IEC 61620
6	Breakdown voltage: - As delivered - After treatment	kV	> 30 > 70	IEC 60156
7	Acidity	mg.KOH/g	< 0.01	IEC 62021
8	Appearance	Clear free from sediment and suspended matter		
9	Corrosive sulphur	-----	Non-corrosive	IEC 62535, dIN 51353
10	Water content	mg / kg	< 30	IEC 60814
11	Oxidation stability (duration 500 h) - Total acidity - Sludge content - Dissipation factor at 90 °C	mg.KOH/g % mass ----	< 0.3 < 0.05 < 0.05	IEC 61125 c IEC 60247
12	PCA	% mass	< 3	IP 346
13	PCB	mg / kg	Not detectable	IEC 61619



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5- TESTING

- Tests shall be carried out in accordance with the latest relevant standard and as specified herein.
- Tests shall be carried out in the presence and under the control of the inspecting engineer of __ __ EDC, except in specified or approved cases where test certificates are accepted. All test results shall be reviewed and approved in writing by __ __ EDC.
- All tests shall be carried out by and at the expense of the tenderer.

6- PACKING and MARKING

- The oil shall be delivered in clean oil drums of appropriate capacity.
- Each oil drum to seal as tight as possible, so that the drum is fully waterproof.
- Each oil drum to be indelibly marked with the following information:
 - Supplier's name
 - Date of drumming
 - Class of Oil
 - Drum Capacity (liters)
 - Filled drum weight kg.
 - Purchase Order No.

7- TECHNICAL DATA SCHEDULE

- Tenderer must declare all additives in the offered oil.
- Tenderer must submit with the offer original catalogue containing full technical data and instructions.
- The vendor guarantee the offered oil against all manufacturing defects.
- The tenderer must fill in thoroughly the attached technical data schedule.
- Any offer does not accompanied with clear and complete technical data schedule shall be rejected.



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Technical Data Schedule

Item	Specification	Unit	Specified Values	Vendor Offered Values
1	Viscosity at 40 °C	mm ² / s	≤ 12	
2	Flash point	°C	> 140	
3	Pour point	°C	≤ -30	
4	Density at 20 °C	kg / m ³	≤ 895	
5	Dissipation factor at 90 °C	---	< 0.005	
6	Breakdown voltage: - Before treatment - After treatment	kV	> 30 > 70	
7	Acidity	mg.KOH/g	< 0.01	
8	Appearance			
9	Corrosive Sulphur	-----	Non-corrosive	
10	Water content	mg / kg	< 30	
11	Oxidation stability (duration 500 h) - Total acidity - Sludge content - Dissipation factor at 90 °C	mg.KOH/g % mass ----	< 0.3 < 0.05 < 0.05	
12	PCA	% mass	< 3	
13	PCB	mg / kg	Not detectable	
14	Carbon type composition	C _P C _A C _N	% mass % mass % mass	

I / We guarantee the correctness of technical data given above for material/equipment we are offering to Electricity Distribution Company.

Signature:

Date: